Cartier–Miller evaluation of genus-one coefficient sums

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Abstract

Let $p \geq 5$ be prime and $f \in \mathbb{F}_p[x]$ a squarefree cubic or quartic. Building on the differentiated Miller chains of Semaev and Rück, we show that, once the trace of Frobenius of $v^2 = f(x)$ is known, one chain of $O(\log p)$ field operations on a quadratic twist evaluates the truncation $T_f = f^{(p-1)/2} \bmod x^p$ at any $\mu \in \mathbb{F}_p^\times$ with $f(\mu) \neq 0$, and the reciprocal character sum $\sum_{x \neq a} \chi(f(x))/(x-a) \in \mathbb{F}_p$ at any a with $f(a) \neq 0$. The key input is an identity valid for every elliptic curve $y^2 = W(x)$ over $\mathbb{F}_p$ and every affine rational point (a, b): b times the high block of $W^{(p-1)/2}$ at a equals the constant Laurent coefficient of the logarithmic derivative of the Miller function of order $\#E(\mathbb{F}_p)$. As an application, for $p \equiv 2 \pmod 3$ the residue $\sum_{k=0}^{(p-2)/3} \binom{2k}{k} 8^{-k} \bmod p$ is computed deterministically in $O(\log p)$ field operations, whereas computing the next partial sum is equivalent to computing $((p-2)/3)! \bmod p$. The method also yields explicit congruences at small-order torsion points, including an evaluation of $\sum_{k=0}^{\lfloor p/3 \rfloor} \binom{2k}{k} \bmod p$. We also interpret the truncation as a logarithmic invariant of a two-pole divisor, normalized directly at the exceptional traces ± 1.

1 Introduction

1.1 The problem

Let $p \geq 5$ be prime, put $h = (p-1)/2$, and let $f \in \mathbb{F}_p[x]$ be squarefree of degree $d \in \{3, 4\}$. Write

$$F = f^h = T_f + x^p J_f, \qquad \deg T_f < p. \tag{1.1}$$

We call T_f the *truncation* and J_f the *high block*. Coefficients of F are classical objects: $[x^{p-1}]F$ is the Hasse invariant of the curve $v^2 = f(x)$ and is congruent to its trace of Frobenius modulo p. This paper is about *values* of T_f and J_f at points of $\mathbb{F}_p$. They are linked to a character sum by two identities (Lemma 2.1): for $a \in \mathbb{F}_p$, with χ the quadratic character,

$$T_f(a) = \chi(f(a)) - aJ_f(a), \qquad C_f(a) := \sum_{x \in \mathbb{F}_p,\, x \neq a} \frac{\chi(f(x))}{x-a} = -F'(a) - J_f(a) \quad \text{in } \mathbb{F}_p. \tag{1.2}$$

For $a \neq 0$ the three values $T_f(a)$, $J_f(a)$ and $C_f(a)$ therefore determine one another with $O(\log p)$ field operations, so evaluating the truncation, the high block or the reciprocal character sum modulo p is one problem. At $a = 0$ both identities still hold, but $T_f(0) = \chi(f(0))$ carries no information about $J_f(0)$; there only $J_f(0)$ and $C_f(0)$ are equivalent. Expanding F costs $O(p)$ field operations, and block-product methods for polynomial recurrences reduce this to $\widetilde{O}(\sqrt{p})$ (Remark 4.5). We show that $O(\log p)$ operations suffice once the trace of Frobenius is known.

Truncations of this kind arise naturally. For $f = 1 - x^k/2$ with $k \in \{3, 4\}$, the value $T_f(1)$ is the partial sum $\sum_{i=0}^{\lfloor (p-1)/k \rfloor} \binom{2i}{i} 8^{-i}$ (Section 6); truncated sums of central binomial coefficients modulo p are a classical topic (Cosgrave & Dilcher, 2010; Sun, 2011). On a short Weierstrass curve of trace one, the function $yJ_W(x)$ is, up to the normalization discussed in Remark 3.3, the function in Voloch's explicit p-descent (Voloch, 1990).

1.2 Main results

Fix a short model $E : y^2 = W(x) = x^3 + Ax + B$ over $\mathbb{F}_p$ with origin O at infinity and uniformizer $t = -x/y$ at O. For a point P and an integer $m \geq 1$, let $f_{m,P}$ be a Miller function, $\operatorname{div}(f_{m,P}) = m(P) - ([m]P) - (m-1)(O)$, and let

$$\alpha_m(P) = [t^0]\,\frac{d\log f_{m,P}}{dt} \tag{1.3}$$

be the constant Laurent coefficient of its logarithmic derivative at O. It does not depend on the scalar normalization of $f_{m,P}$, and Miller's addition chain computes it in $O(\log m)$ field operations (Algorithm 2.2).

Theorem A (Curve-order identity)**.** *Let $N = \#E(\mathbb{F}_p)$ and write $W^h = U + H_E x^{p-1} + x^p J_W$ with $\deg U \leq p-2$. For every affine point $P = (a,b) \in E(\mathbb{F}_p)$,*

$$\alpha_N(P) = b\,J_W(a).$$

No hypothesis on N is needed: N may be divisible by p, and E may be supersingular.

When $f(a)$ is a square there is a rational point above a and Theorem A applies to a model of $v^2 = f(x)$ directly. In general one normalizes at a, which moves to the quadratic twist by $f(a)$, where the point above a becomes rational.

Theorem B (Evaluation at a nonbranch point)**.** *Let $a \in \mathbb{F}_p$ with $f(a) \neq 0$, put $\varepsilon = \chi(f(a))$, let τ_f be the trace of Frobenius of the smooth projective curve $v^2 = f(x)$, and put $N = p + 1 - \varepsilon\tau_f$. There is an explicit short Weierstrass curve E_a, a model of $v^2 = f(a+z)/f(a)$ with $\#E_a(\mathbb{F}_p) = N$, and an explicit point $R \in E_a(\mathbb{F}_p)$, both given by polynomials in the Taylor coefficients of f at a (equation (4.2)), such that, with $g_1 = f'(a)/f(a)$,*

$$C_f(a) = \varepsilon\Big(-\frac{g_1}{4}(N-2) - \frac{\alpha_N(R)}{2}\Big), \qquad T_f(a) = \varepsilon\Big(1 - \frac{aNg_1}{4} - \frac{a}{2}\,\alpha_N(R)\Big)\ (a \neq 0).$$

Corollary C (Complexity)**.** *Given τ_f, at each nonbranch point a the values $C_f(a)$, $J_f(a)$ and, for $a \neq 0$, $T_f(a)$ cost $O(\log p)$ field operations and a constant number of field registers. Since τ_f can be computed deterministically in time polynomial in $\log p$ (Schoof, 1985), these values are computable deterministically in time polynomial in $\log p$; further points on the same curve reuse τ_f.*

Section 5 explains where the formula comes from. For $f(0) = 1$ and $\lambda = 1/\mu$, the value $T_f(\mu)$ is the coefficient $[w^{p-1}]f(w)^h/(1-\lambda w)$, and Theorem D identifies it with a logarithmic invariant of the two-pole divisor $(\mu, v_0) - (\mu, -v_0)$, $v_0^2 = f(\mu)$. When the trace equals $\varepsilon = \pm 1$ the natural annihilator of that divisor is p itself, and the identity requires a separate normalization (Lemma 5.2).

1.3 A binomial example

Put $a_k = \binom{2k}{k}8^{-k}$ and $B_K = \sum_{k=0}^{K-1} a_k$ in $\mathbb{F}_p$. For $f = 1 - x^3/2$ the coefficient of x^{3k} in f^h is a_k for $0 \leq k \leq h$, and all other coefficients vanish, so for $p \equiv 2 \pmod 3$ and $L = (p+1)/3$ the truncation gives $T_f(1) = B_L$. The curve $v^2 = 1 - x^3/2$ is supersingular for these primes, so its trace is 0 and no point counting is needed.

Corollary E (A partial sum and its next term)**.** *Let $p \geq 5$ with $p \equiv 2 \pmod 3$, and put $L = (p+1)/3$ and $m = L - 1 = (p-2)/3$.*

(i) *The residue $B_L = \sum_{k=0}^{m} \binom{2k}{k}8^{-k} \bmod p$ is computable deterministically in $O(\log p)$ field operations.*

(ii) *The next term satisfies* $a_L = 3/(2\,m!^3)$ *and* $m! = (3/(2a_L))^{(2p-1)/3}$ *in* $\mathbb{F}_p$. *Hence computing* $B_{L+1} = B_L + a_L$ *is equivalent, up to* $O(\log p)$ *field operations, to computing* $((p-2)/3)! \bmod p$.

The equivalence in (ii) is a reduction in both directions; it proves no lower bound for either task and no complexity separation between B_L and B_{L+1}. We know of no polynomial-time algorithm for $((p-2)/3)! \bmod p$. Classical deterministic methods use $\widetilde{O}(\sqrt{p})$ field operations (Bostan, Gaudry & Schost, 2007). Tal (2026) goes below the square-root exponent by Jacobi-sum reconstruction when $p-1$, or another cyclotomic value $\Phi_r(p)$, has a divisor q in a suitable range: deterministically in $\widetilde{O}(q^c + \sqrt{p/q})$ bit operations for a supplied $q \mid p-1$, and with expected bounds (Las Vegas setup) in the cyclotomic variants, which together cover all but an arbitrarily small proportion of primes. For arbitrary p its deterministic bound improves the classical one by a logarithmic factor. This is the relevant route to boundary terms such as a_L. In this example, therefore, the partial sum up to the cutoff is a value of a truncated power on a curve, while one more term is a modular factorial. At the quarter point ($p \equiv 1 \pmod 4$, $f = 1 - x^4/2$) both the prefix and the boundary term are accessible, the latter through Gauss's binomial congruence (Section 6). Section 6.3 shows that when the point R of Theorem B is a torsion point the evaluator gives closed forms; for example $\sum_{k=0}^{\lfloor p/3 \rfloor} \binom{2k}{k} \equiv -\frac{1}{3}$ or $\frac{1-2r}{3} \pmod p$ according as $p \equiv 2$ or $1 \pmod 3$, where $4p = r^2 + 27s^2$ (Theorem 6.3).

1.4 Relation to previous work

The ingredients are classical. Miller (1986, 2004) introduced functions $f_{m,P}$ with prescribed divisors and their evaluation along an addition chain. Semaev (1998) differentiated Miller's recurrence to compute the logarithmic-derivative map on the p-torsion of an anomalous curve in $O(\log p)$ operations, and Rück (1999) extended this to divisor class groups, making the local-coefficient cocycle explicit; see also Galbraith (2012, Section 26.4.1). Differentiated Miller evaluation, and the cocycle in Lemma 3.1, belong to this Semaev–Rück framework. The Cartier operator and its action on logarithmic differentials go back to Cartier (1957) and Serre (1958). Voloch (1990, 1997) identified the Artin–Schreier construction behind the high block with Verschiebung and obtained explicit p-descent; on a trace-one curve the relevant function is $yJ_W(x)$ up to normalization, and Theorem A for $N = p$ identifies it with the Miller coefficient α_p in an explicit normalization (Remark 3.3). Cassels (1971) and Gunji (1976) studied the related logarithmic differentials and quasi-periods. The anomalous case is the setting of the discrete-logarithm attacks of Satoh and Araki (1998) and Smart (1999). For trace-free computation, the block-product methods of Bostan, Gaudry and Schost (2007) and Harvey (2007) apply, and Harvey and Sutherland (2014, 2016) treat many primes simultaneously.

The proposed contribution is specific: (i) the curve-order identity of Theorem A for every N, including $p \nmid N$, where $d \log f_{N,P}$ has a pole at O and its Laurent constant is used; (ii) the twist normalization that turns it into a base-field evaluator at every nonbranch point, with the weighted-sum bookkeeping of Section 4; (iii) the normalization of the residue-divisor identity at both exceptional traces; and (iv) the binomial consequences of Section 6. We have not found these statements in the literature we examined; priority remains subject to specialist assessment, and we would welcome references to earlier appearances.

1.5 Organization and code

Section 2 collects notation and the Miller recurrence. Theorem A is proved in Section 3, and Theorem B with its corollaries in Section 4. Section 5 gives the residue-divisor interpretation, Section 6 the binomial applications and Section 7 numerical checks and open questions. A short script that checks the identities and reproduces the numbers in this paper, its saved output, and a computational companion on boundary acquisition and benchmarks (Jordan, 2026), is

available at https://github.com/bbrhuft/Cartier-Miller-evaluation-of-genus-one-coefficient-sums.

2 Preliminaries

2.1 Notation

Throughout, $p \geq 5$ and $h = (p-1)/2$. The quadratic character is $\chi : \mathbb{F}_p \to \{0, \pm 1\}$ with $\chi(0) = 0$, and $\varepsilon_f(x) = \chi(f(x))$; a single character value is written ε. For a squarefree f of degree $d \in \{3, 4\}$ with leading coefficient ℓ_f, let $N_\infty(f)$ be the number of rational points at infinity of the smooth projective model of $v^2 = f(x)$: $N_\infty(f) = 1$ if $d = 3$ and $1 + \chi(\ell_f)$ if $d = 4$. Let $\tau_f = p + 1 - \#\{\text{rational points}\}$ be its trace. Counting fibres gives

$$\sum_{x \in \mathbb{F}_p} \varepsilon_f(x) = 1 - N_\infty(f) - \tau_f. \tag{2.1}$$

Frobenius on points is π; Verschiebung is Ver. Power sums over $\mathbb{F}_p$ satisfy $\sum_x x^k = -1$ if $k > 0$ and $(p-1) \mid k$, and 0 otherwise.

2.2 The Cartier operator

On a smooth curve over a perfect field of characteristic p, with a separating coordinate x, every differential has a unique expression $\sum_{i=0}^{p-1} u_i^p \, x^i \, dx$, and the Cartier operator is $\mathcal{C}(\eta) = u_{p-1} \, dx$. It is p^{-1}-linear, $\mathcal{C}(a^p \eta) = a \, \mathcal{C}(\eta)$, it kills exact differentials, and it fixes logarithmic differentials dg/g (Cartier, 1957; Serre, 1958). On Laurent or polynomial expressions in x,

$$\mathcal{C}\Big(\sum_n b_n x^n \, dx\Big) = \sum_j b_{pj+p-1}^{1/p} \, x^j \, dx. \tag{2.2}$$

2.3 Truncation, high block and character sum

Lemma 2.1. *Let f be a squarefree cubic or quartic, $F = f^h = T_f + x^p J_f$ as in (1.1), and $a \in \mathbb{F}_p$. Then*

$$T_f(a) = \varepsilon_f(a) - a J_f(a), \qquad C_f(a) = -F'(a) - J_f(a), \tag{2.3}$$

where $F'(a) = -\varepsilon_f(a) f'(a)/(2f(a))$ if $f(a) \neq 0$ and $F'(a) = 0$ if $f(a) = 0$. Moreover $[x^{p-1}]F \equiv \tau_f \pmod p$.

Proof. Since $a^p = a$, Euler's criterion gives $\varepsilon_f(a) = F(a) = T_f(a) + a J_f(a)$. For the second identity put $q(x) = (F(x) - F(a))/(x - a)$, a polynomial of degree at most $2p - 3$. As $\sum_{x \neq a} (x-a)^{-1} = 0$ and $F(x) = \varepsilon_f(x)$ on $\mathbb{F}_p$, we get $C_f(a) = \sum_{x \in \mathbb{F}_p} q(x) - q(a)$. The power sums give $\sum_x q(x) = -[x^{p-1}]q$, and expanding $(x^k - a^k)/(x - a)$ shows $[x^{p-1}]q = \sum_{k \geq p} [x^k]F \cdot a^{k-p} = J_f(a)$. Finally $q(a) = F'(a) = h f'(a) f(a)^{h-1}$; at a nonbranch point this is $-\varepsilon_f(a) f'(a)/(2f(a))$ because $h = -1/2$ in $\mathbb{F}_p$, and at a branch point it vanishes because $h \geq 2$.

For the Hasse invariant, $\sum_x \varepsilon_f(x) = \sum_x F(x) = -[x^{p-1}]F - [x^{2p-2}]F$, and $[x^{2p-2}]F = \ell_f^h = \chi(\ell_f)$ when $d = 4$ and is 0 when $d = 3$. Comparing with (2.1) gives $[x^{p-1}]F \equiv \tau_f$. □

Given $\varepsilon_f(a)$ and $F'(a)$, which cost $O(\log p)$ operations, (2.3) converts any one of $T_f(a)$, $J_f(a)$, $C_f(a)$ into the others when $a \neq 0$. At $a = 0$ it relates only $J_f(0)$ and $C_f(0)$, since $T_f(0) = \varepsilon_f(0)$.

2.4 Miller coefficients

On $E : y^2 = x^3 + Ax + B$ use $t = -x/y$. The invariant differential $\omega_0 = dx/(2y)$ satisfies $\omega_0 = (1 + O(t^4))\,dt$. A nonvertical line of slope ρ and a vertical line have expansions

$$y - \rho x - \nu = -t^{-3}\big(1 + \rho t + O(t^3)\big), \qquad x - c = t^{-2}\big(1 - ct^2 + O(t^4)\big), \tag{2.4}$$

so the constant term of the logarithmic derivative is ρ for the line and 0 for the vertical. Since $f_{i+j,P} = f_{i,P}\, f_{j,P}\, \ell/v$ up to a constant, where ℓ is the line through $[i]P, [j]P$ and v the vertical through $[i + j]P$ (Miller, 2004; Galbraith, 2012, Section 26.3.1), the coefficients (1.3) obey

$$\alpha_{2i}(P) = 2\alpha_i(P) + \rho, \qquad \alpha_{i+1}(P) = \alpha_i(P) + \rho, \tag{2.5}$$

with ρ the slope used at that step, and $\rho = 0$ for vertical and identity steps.

Algorithm 2.2 (Differentiated Miller chain)**.** Given $P \in E(\mathbb{F}_p)$ and $m \geq 1$, set $T \leftarrow P$, $\alpha \leftarrow 0$. For each binary digit of m after the leading 1: replace (T, α) by $([2]T,\ 2\alpha + \rho)$, where ρ is the tangent slope at T; if the digit is 1, replace (T, α) by $(T + P,\ \alpha + \rho)$, where ρ is the chord slope. Return $\alpha = \alpha_m(P)$ and $T = [m]P$.

Conventions: the slope is determined by the operands (x_1, y_1) and (x_2, y_2), not by the type of step. If either operand is O, the step returns the other operand and $\rho = 0$. If $x_1 = x_2$ and $y_1 = -y_2$ (including doubling a point of order two), it returns O and $\rho = 0$. Otherwise, if the operands coincide ($x_1 = x_2$, $y_1 = y_2 \neq 0$), ρ is the tangent slope $(3x_1^2 + A)/(2y_1)$, and if $x_1 \neq x_2$, ρ is the chord slope $(y_2 - y_1)/(x_2 - x_1)$. In particular an addition step can meet $T = P$, which happens when the order of P is smaller than m, and then uses the tangent. The doubling update $\alpha \leftarrow 2\alpha + \rho$ is applied even when $T = O$.

The chain uses $O(\log m)$ field operations, each step including one inversion, and a constant number of registers.

3 The curve-order identity

Let $E : y^2 = W(x) = x^3 + Ax + B$ over $\mathbb{F}_p$ be nonsingular, $N = \#E(\mathbb{F}_p) = p + 1 - \tau_E$, and write

$$W^h = U + H_E x^{p-1} + x^p J_W, \qquad \deg U \leq p - 2. \tag{3.1}$$

By Lemma 2.1, $H_E \equiv \tau_E \pmod p$. For geometric points S, T let $g_{S,T}$ be the function with divisor $(S) + (T) - (S + T) - (O)$ (a line over a vertical, a vertical, or 1) and $c(S, T) = [t^0]\, d \log g_{S,T}/dt$.

Lemma 3.1. *For geometric points S, T, Q and integers $m, n \geq 1$,*

$$\alpha_m(S + T) = \alpha_m(S) + \alpha_m(T) + c([m]S, [m]T) - m\, c(S, T), \tag{3.2}$$

$$\alpha_{mn}(Q) = n\, \alpha_m(Q) + \alpha_n([m]Q), \tag{3.3}$$

and $\alpha_m(-S) = -\alpha_m(S)$. In particular α_p is additive when one summand is p-torsion.

Proof. Both sides of each identity are constant terms of logarithmic derivatives of functions with the same divisor: for (3.2), $f_{m,S+T}\, g_{S,T}^m$ and $f_{m,S} f_{m,T}\, g_{[m]S,[m]T}$; for (3.3), $f_{mn,Q}$ and $f_{m,Q}^n f_{n,[m]Q}$. Negation $(x, y) \mapsto (x, -y)$ sends t to $-t$ and $f_{m,S}$ to a function with the divisor of $f_{m,-S}$, which gives the sign rule. If $[p]T = O$, then $c([p]S, [p]T) = c([p]S, O) = 0$ and $p\, c(S, T) = 0$ in characteristic p. □

Theorem 3.2 (Theorem A)**.** *For every affine $P = (a, b) \in E(\mathbb{F}_p)$, $\alpha_N(P) = b\, J_W(a)$.*

Proof. The case $b = 0$. Then P has order two, N is even, and $f_{N,P}$ is a constant multiple of $(x-a)^{N/2}$, whose logarithmic derivative has constant term 0 by (2.4).

A Cartier computation. Let $b \neq 0$ and consider the third-kind differential

$$\vartheta_P = \frac{y+b}{x-a}\,\omega_0 = \frac{1}{2}\,\frac{dx}{x-a} + \frac{b}{2}\,\frac{dx}{(x-a)y}.$$

It has simple poles at P and O with residues $+1$ and -1 and is regular elsewhere, including at $-P$. The first summand is logarithmic, hence fixed by $\mathcal{C}$. For the second, write $1/y = W^h/y^p$ and $W^h/(x-a) = q(x) + 1/(x-a)$ with $q = (W^h-1)/(x-a)$; here $W(a)^h = b^{p-1} = 1$, so q is a polynomial of degree less than $2p-1$. By (2.2), $\mathcal{C}(q\,dx) = q_{p-1}\,dx$ with $q_{p-1} = [x^{p-1}]q = J_W(a)$, as in the proof of Lemma 2.1. Hence

$$(\mathcal{C}-1)\,\vartheta_P = b\,J_W(a)\,\omega_0. \tag{3.4}$$

Comparison with a Miller function. Choose a geometric point Q with $[p]Q = P$, put $\beta = \alpha_p(Q)$ and $\theta = -d\log f_{p,Q}$. Since $\operatorname{div} f_{p,Q} = p(Q) - (P) - (p-1)(O)$, the differential θ has the same residues as ϑ_P and is fixed by $\mathcal{C}$. At O, $\vartheta_P/\omega_0 = -t^{-1} + O(t)$ and $\theta/\omega_0 = -t^{-1} - \beta + O(t)$, so the holomorphic difference is $\vartheta_P - \theta = \beta\,\omega_0$. Since $\mathcal{C}\omega_0 = H_E^{1/p}\omega_0$ by (2.2), applying $\mathcal{C}-1$ to this difference and using (3.4) gives $bJ_W(a) = (H_E\beta)^{1/p} - \beta$; as $bJ_W(a) \in \mathbb{F}_p$, raising to the pth power gives

$$b\,J_W(a) = H_E\,\beta - \beta^p. \tag{3.5}$$

Galois descent. From $\pi\operatorname{Ver} = [p]$ and $\pi(P) = P$ we get $\pi(\operatorname{Ver} Q) = \pi(P)$, so $\operatorname{Ver} Q = P$ because π is injective on geometric points. The relation $\pi + \operatorname{Ver} = [\tau_E]$ then gives $\pi(Q) = [\tau_E - p]Q = Q - [N]Q$. Put $T = [N]Q$, so that $[p]T = [N]P = O$. Since E and the uniformizer are defined over $\mathbb{F}_p$, $\alpha_p(Q)^p = \alpha_p(\pi Q)$, and Lemma 3.1 gives

$$\beta^p = \alpha_p(Q-T) = \beta - \alpha_p(T), \qquad N\beta + \alpha_N(P) = \alpha_{pN}(Q) = p\,\alpha_N(Q) + \alpha_p(T) = \alpha_p(T),$$

using (3.3) in both orders. Substituting into (3.5) yields $bJ_W(a) = (H_E - 1 + N)\beta + \alpha_N(P)$, and $H_E - 1 + N \equiv \tau_E - 1 + 1 - \tau_E = 0 \pmod p$. □

The point Q is a device of the proof and is never computed.

Remark 3.3. (a) At trace one, $N = p$ and the theorem identifies $\alpha_p(P)$ with the value at P of $yJ_W(x)$, in the normalization fixed by $t = -x/y$ and $\omega_0 = dx/(2y)$. For the representative $(P) - (-P)$ the corresponding scalar is $2bJ_W(a)$. Comparison with Voloch (1990) needs care: Voloch (1997, Section 3), crediting A. Broumas, records that the 1990 paper used dx/y where $dx/(2y)$ is correct, so that a factor -2 is missing throughout it, and that its z is $-z$ in the 1997 normalization. Scalars from the 1990 paper must be adjusted accordingly. Our normalization does not rely on that comparison; Section 7 checks it by constructing $f_{N,P}$ without Algorithm 2.2, including on trace-one curves. (b) If $p \nmid N$, the rational point $Q = [p^{-1} \bmod N]P$ is a valid choice in the proof and gives $bJ_W(a) = (H_E - 1)\alpha_p(Q)$. Theorem A avoids this modular inverse and the preliminary scalar multiplication. (c) When $p \nmid N$, $d\log f_{N,P}$ has residue $-N \neq 0$ at O, so $\alpha_N(P)$ is a Laurent constant and not the value of a holomorphic differential. The map $P \mapsto \alpha_N(P)$ is not claimed to be a homomorphism on $E(\mathbb{F}_p)$.

4 Evaluation at nonbranch points and weighted sums

In this section f is any squarefree cubic or quartic over $\mathbb{F}_p$, $p \geq 5$, with no condition on its constant term.

4.1 The normalized twist

Let $f(a) \neq 0$, $\varepsilon = \varepsilon_f(a)$, and write

$$g(z) = \frac{f(a+z)}{f(a)} = 1 + g_1 z + g_2 z^2 + g_3 z^3 + g_4 z^4, \qquad g_1 = \frac{f'(a)}{f(a)}, \tag{4.1}$$

with $g_4 = 0$ for a cubic. The curve $v^2 = g(z)$ has the rational points $(0, \pm 1)$. Taking $(0,1)$ as origin, the model change of Appendix A gives the short model $E_a : Y^2 = X^3 + A_g X + B_g$, and the image R of $(0,-1)$, where

$$\begin{aligned} A_g &= -4g_4 + g_1 g_3 - \tfrac{1}{3} g_2^2, \\ B_g &= g_3^2 - \tfrac{8}{3} g_2 g_4 + g_1^2 g_4 - \tfrac{1}{3} g_1 g_2 g_3 + \tfrac{2}{27} g_2^3, \\ R &= (x_R, y_R) = \Big(\tfrac{1}{4} g_1^2 - \tfrac{2}{3} g_2,\ -g_3 + \tfrac{1}{2} g_1 g_2 - \tfrac{1}{8} g_1^3 \Big). \end{aligned} \tag{4.2}$$

Since $\varepsilon_f(a+z) = \varepsilon\, \varepsilon_g(z)$, and since for a quartic the normalization multiplies the leading coefficient by $1/f(a)$, (2.1) gives $\tau_g = \varepsilon \tau_f$; hence $\#E_a(\mathbb{F}_p) = N := p + 1 - \varepsilon \tau_f$.

Theorem 4.1 (Theorem B)**.** *With this notation,*

$$C_f(a) = \varepsilon \Big(-\frac{g_1}{4}(N-2) - \frac{\alpha_N(R)}{2} \Big). \tag{4.3}$$

Proof. Since $C_f(a) = \sum_{z \neq 0} \varepsilon_f(a+z)/z = \varepsilon\, C_g(0)$, it suffices to compute $C_g(0)$. From the formulas (A.1) one checks the identity of rational functions

$$\frac{1}{z} = \frac{1}{2}\, \varphi_R - \frac{g_1}{4}, \qquad \varphi_R = \frac{Y + y_R}{X - x_R}, \tag{4.4}$$

where removable singularities take their limiting values. The function $1/z$ vanishes at the points at infinity of $v^2 = g(z)$, and its sum over the affine points with $z \neq 0$ is $\sum_{z \neq 0} (1 + \varepsilon_g(z))/z = C_g(0)$. The isomorphism sends $(0,1) \mapsto O$ and $(0,-1) \mapsto R$, so summing (4.4) over $S \in E_a(\mathbb{F}_p) \setminus \{O, R\}$, a set of $N-2$ points, gives

$$C_g(0) = \frac{1}{2} \sum_{S \neq O, R} \varphi_R(S) - \frac{g_1}{4}(N-2).$$

It remains to show $\sum_{S \neq O,R} \varphi_R(S) = -\alpha_N(R)$. Suppose $y_R \neq 0$ and let $W_g(X) = X^3 + A_g X + B_g$. At $-R$ the function $\varphi_R = (W_g(X) - W_g(x_R))/((X - x_R)(Y - y_R))$ has value $-W_g'(x_R)/(2y_R)$. Over the abscissae $X \neq x_R$, the terms $Y/(X - x_R)$ cancel in pairs $\pm Y$, so these points contribute $y_R \sum_{X \neq x_R} (1 + \varepsilon_{W_g}(X))/(X - x_R) = y_R\, C_{W_g}(x_R)$. Lemma 2.1 on the short model, where $\varepsilon_{W_g}(x_R) = \chi(y_R^2) = 1$, gives

$$\sum_{S \neq O, R} \varphi_R(S) = y_R C_{W_g}(x_R) - \frac{W_g'(x_R)}{2y_R} = -y_R\, J_{W_g}(x_R) = -\alpha_N(R),$$

the last step by Theorem 3.2. If $y_R = 0$, negation pairs the terms $Y/(X - x_R)$, the fixed points contribute zero, and $\alpha_N(R) = 0$ by Theorem 3.2. □

Corollary 4.2. *If moreover $a \neq 0$, then*

$$T_f(a) = \varepsilon \Big(1 - \frac{aN f'(a)}{4 f(a)} - \frac{a}{2}\, \alpha_N(R) \Big). \tag{4.5}$$

Proof. By Lemma 2.1, $T_f(a) = \varepsilon - a J_f(a) = \varepsilon + a C_f(a) + a F'(a)$ with $F'(a) = -\varepsilon f'(a)/(2f(a))$. Substituting (4.3) combines the two derivative terms into $-\varepsilon a N f'(a)/(4f(a))$. □

Theorem B includes $p = 5$, where N may equal $2p$; for example it returns $C_{x^3+3x}(1) = 1$ over $\mathbb{F}_5$, where $\tau_f = -4$. Zero constant terms and the pole $a = 0$ need no special treatment. Every pole is used as its own normalization origin, and the trace τ_f, computed once, supplies N at every pole through the sign ε.

Corollary 4.3. *Suppose the exact trace τ_f is supplied. At a nonbranch point a, each of $C_f(a)$, $J_f(a)$ and, for $a \neq 0$, $T_f(a)$ costs $O(\log p)$ field operations, including $O(\log p)$ inversions, with a constant number of field registers; with schoolbook arithmetic this is $O((\log p)^3)$ bit operations. Since Schoof's algorithm computes τ_f deterministically in time polynomial in $\log p$ (Schoof, 1985, 1995), the values are computable deterministically in polynomial time.*

Proof. Forming g, A_g, B_g and R costs $O(1)$ operations apart from one exponentiation for ε; all denominators in (4.2) are units for $p \geq 5$. By Hasse's bound $N = O(p)$, so the chain has $O(\log p)$ steps. To apply Schoof's algorithm, normalize at any nonroot of f among $\deg f + 1$ fixed elements to obtain a short model, and recover τ_f from its trace by the sign rule. □

A consistency check comes free: the chain returns $[N]R$, which must be O. This detects most wrong traces but does not certify the trace.

4.2 Several poles and complete excluded domains

Theorem 4.4. *Let $\mathcal{A} = \{a_1, \dots, a_s\} \subset \mathbb{F}_p$ be distinct, $\Omega = \mathbb{F}_p \setminus \mathcal{A}$, and $\mathcal{R}(x) = r_0 + \sum_i r_i/(x - a_i)$, with zero residues allowed. Put $w_i = \varepsilon_f(a_i)$, $K_f = \sum_x \varepsilon_f(x) = 1 - N_\infty(f) - \tau_f$ and $D_i(v) = \sum_{j \neq i} v_j/(a_i - a_j)$. Then*

$$\sum_{x \in \Omega} \mathcal{R}(x)\varepsilon_f(x) = r_0\Big(K_f - \sum_i w_i\Big) + \sum_i r_i\big(C_f(a_i) + D_i(w)\big), \tag{4.6}$$

$$\sum_{\substack{P \text{ affine} \\ x(P) \in \Omega}} \mathcal{R}(x(P)) = r_0\Big(K_f - s - \sum_i w_i\Big) + \sum_i r_i\big(C_f(a_i) + D_i(1 + w)\big). \tag{4.7}$$

Proof. $C_f(a_i)$ excludes only a_i; removing each other a_j subtracts $w_j/(a_j - a_i)$, which is the correction $D_i(w)$. For the point sum, the fibre over x has $1 + \varepsilon_f(x)$ points, $\sum_{x \in \Omega}(x - a_i)^{-1} = D_i(1)$, and $|\Omega| = p - s \equiv -s$. □

The identities (4.6)–(4.7) are algebraic and hold for every excluded set, including branch points with nonzero residue, when the exact values $C_f(a_i)$ are used. The fast evaluation obtains $C_f(a_i)$ from Theorem B and therefore needs the hypothesis

$$r_i \neq 0 \implies f(a_i) \neq 0.$$

Under this hypothesis, with the trace supplied, (4.6) costs $O(s \log p + s^2)$ field operations: one chain per active pole, one character evaluation per excluded point, and the pairwise corrections. An excluded branch point with zero residue remains allowed, because its oracle value is not needed; it enters only through $w_i = 0$. The quadratic term can be reduced with product and remainder trees: with $B(X) = \prod_j (X - a_j)$ and $W_v(X) = \sum_j v_j B(X)/(X - a_j)$,

$$D_i(v) = \frac{W_v'(a_i)}{B'(a_i)} - \frac{v_i B''(a_i)}{2B'(a_i)}. \tag{4.8}$$

The projective point sum adds $r_0 N_\infty(f)$ to (4.7).

Remark 4.5 (A trace-free alternative)**.** Let $d = \deg f$, $Q_a(z) = z^d f(a + 1/z)$ and $k = dh - p$. Since $(a + z)^p = a + z^p$ and $\deg J_f < p$, the coefficient of z^p in $F(a + z)$ is $J_f(a)$; reversing gives

$$J_f(a) = [z^k]\, Q_a(z)^h, \tag{4.9}$$

valid at every a, including branch points. The coefficients t_k of $Q_a^h = \sum t_k z^k$ satisfy the recurrence $k\, g_0 t_k = \sum_{j=1}^{\min(d,k)} ((h+1)j - k)\, g_j t_{k-j}$, where $Q_a = \sum g_j z^j$ and $g_0 = \ell_f \neq 0$. This is a linear recurrence with polynomial coefficients, so the block-product methods of Bostan, Gaudry and Schost (2007) and Harvey (2007, Theorem 10) evaluate (4.9) in $\widetilde{O}(\sqrt{p})$ field operations without a trace. The trace-supplied and trace-free models are not directly comparable; the companion report compares implementations of both.

5 The residue-divisor interpretation

This section explains Theorem B in terms of the Cartier operator on the original curve, and records the exceptional normalization. Assume $p \geq 7$ and $f(0) = 1$, let E be the smooth projective curve $v^2 = f(w)$ with origin $O = (0, 1)$ and $\omega = dw/v$, and write $\tau := \tau_f$ for its trace. The restriction $p \geq 7$ is used only to locate the exceptional case via Hasse's bound; the evaluator of Section 4 is valid for $p \geq 5$. For $\lambda \in \mathbb{F}_p^\times$ with $\mu = 1/\lambda$ and $f(\mu) \neq 0$, let $H = [w^{p-1}]f^h$ and put

$$\begin{gathered} S_f(\lambda) = [w^{p-1}]\frac{f(w)^h}{1-\lambda w} = T_f(\mu), \qquad v_0^2 = f(\mu), \qquad \varepsilon = \chi(f(\mu)), \\ Q_\pm = (\mu, \pm v_0), \qquad \mathfrak{D} = (Q_+) - (Q_-), \qquad r = \frac{1}{\lambda v_0}. \end{gathered} \tag{5.1}$$

The second equality in the first formula uses $\lambda^{p-1} = 1$. For an integer M annihilating the class of $\mathfrak{D}$, choose $F_\mathfrak{D}$ with $\operatorname{div} F_\mathfrak{D} = M\mathfrak{D}$ and set $\Psi_M(\mathfrak{D}) = ((dF_\mathfrak{D}/F_\mathfrak{D})/\omega)(O)$; this does not depend on the scalar normalization of $F_\mathfrak{D}$.

Theorem D (Residue-divisor identity)**.** *Under these hypotheses, with $M_\varepsilon = p + 1 - \varepsilon\tau$ and $\tau = \tau_f$,*

$$S_f(\lambda) = H - \varepsilon\, r\, \Psi_{M_\varepsilon}(\mathfrak{D}).$$

The product $r\Psi_{M_\varepsilon}(\mathfrak{D})$ lies in $\mathbb{F}_p$ and is unchanged by $v_0 \mapsto -v_0$. The formula includes the exceptional case $\tau = \varepsilon$, where $M_\varepsilon = p$.

5.1 The regular case

Put $\theta_\lambda = dw/((1-\lambda w)v)$. It has simple poles only at $Q_\pm$, with residues $\mp r$. Since $v^{p-1} = f^h$ and w is a uniformizer at O, (2.2) gives

$$\mathcal{C}(\omega) = H\omega, \qquad \Big(\frac{\mathcal{C}(\theta_\lambda)}{\omega}\Big)(O) = S_f(\lambda)^{1/p}. \tag{5.2}$$

Proposition 5.1. *Let $M \geq 1$ annihilate the class of $\mathfrak{D}$, with $p \nmid M$. Then $M\big(S_f(\lambda) - H\big) = r\big(H\Psi_M(\mathfrak{D}) - \Psi_M(\mathfrak{D})^p\big)$.*

Proof. The residues cancel in $M\theta_\lambda + r\, dF_\mathfrak{D}/F_\mathfrak{D}$, so this differential equals $c\,\omega$ with $c = M + r\Psi_M(\mathfrak{D})$, by evaluation at O. Apply $\mathcal{C}$, which fixes $dF_\mathfrak{D}/F_\mathfrak{D}$, evaluate at O using (5.2), and raise to the pth power: $MS_f(\lambda) + r\Psi_M(\mathfrak{D})^p = (M + r\Psi_M(\mathfrak{D}))H$. □

Proof of Theorem D when $p \nmid M_\varepsilon$. Frobenius maps Q_+ to Q_ε, so $\pi(\mathfrak{D}) = \varepsilon\mathfrak{D}$ as divisors, and the characteristic equation of Frobenius shows that M_ε annihilates the class of $\mathfrak{D}$. Coefficient conjugation sends $F_\mathfrak{D}$ to a constant multiple of $F_\mathfrak{D}^\varepsilon$, so $\Psi_{M_\varepsilon}(\mathfrak{D})^p = \varepsilon\Psi_{M_\varepsilon}(\mathfrak{D})$. Substituting into Proposition 5.1 and using $M_\varepsilon \equiv 1 - \varepsilon H \pmod p$ gives the theorem. Changing the sign of v_0 negates both r and Ψ. □

By Hasse's bound, $|\tau| \leq 2\sqrt{p} < p - 1$ for $p \geq 7$, so $p \mid M_\varepsilon$ exactly when $\tau = \varepsilon$, and then $M_\varepsilon = p$. The class of $\mathfrak{D}$ is nonzero, since distinct points on a genus-one curve have nonprincipal difference, so it has order exactly p, and Proposition 5.1 carries no information.

5.2 The exceptional case

Assume $\tau = \varepsilon \in \{\pm 1\}$ and work over $k = \overline{\mathbb{F}}_p$. For a degree-zero divisor $\mathfrak{D}_0$ with p-torsion class choose $\operatorname{div} F_{\mathfrak{D}_0} = p\mathfrak{D}_0$; its logarithmic differential is holomorphic, and

$$\frac{dF_{\mathfrak{D}_0}}{F_{\mathfrak{D}_0}} = \Psi_p([\mathfrak{D}_0])\,\omega, \qquad \Psi_p : \mathrm{Pic}^0(E)[p](k) \to k \tag{5.3}$$

is an additive, injective map (Serre, 1958, Proposition 10). Since $\mathcal{C}\omega = \varepsilon\omega$, its image lies in the one-dimensional $\mathbb{F}_p$-space $\mathscr{L}_\varepsilon = \{b \in k : b^p = \varepsilon b\}$. Coefficient conjugation σ raises constants to the pth power and fixes the coordinate functions.

Lemma 5.2 (Exceptional Cartier defect)**.** *Suppose $\tau = \varepsilon \in \{\pm 1\}$. Let $\mathfrak{D}_0$ be a degree-zero divisor whose class is killed by p and with $\sigma(\mathfrak{D}_0) = \varepsilon\mathfrak{D}_0$. If $\theta_{\mathfrak{D}_0}$ is a differential over $\mathbb{F}_{p^2}$ with only simple poles, residue divisor $\mathfrak{D}_0$, and $\sigma(\theta_{\mathfrak{D}_0}) = \varepsilon\theta_{\mathfrak{D}_0}$, then*

$$(\mathcal{C} - 1)\,\theta_{\mathfrak{D}_0} = \Psi_p([\mathfrak{D}_0])\,\omega. \tag{5.4}$$

In other words, the Cartier defect and the logarithmic transfer are the same map on geometric p-torsion, with comparison factor exactly 1.

The lemma fixes the scalar in the exceptional case, where Proposition 5.1 is empty; for $\varepsilon = -1$ the natural divisor $(P) - (-P)$ has class $2[(P) - (O)]$, and the lemma accounts for this factor through $\Psi_p([\mathfrak{D}_0])$. Its proof, by descent through Verschiebung, is given in Appendix B.

Proof of Theorem D when $\tau = \varepsilon$. Put $\widehat{\theta} = -\theta_\lambda/r$. It has residue divisor $\mathfrak{D}$, and $\sigma(\widehat{\theta}) = \varepsilon\widehat{\theta}$ because $r^p = \varepsilon r$. Lemma 5.2 gives $(\mathcal{C} - 1)\widehat{\theta} = \Psi_p(\mathfrak{D})\,\omega$. Evaluating at O with (5.2) and using $r^{1/p} = \varepsilon r$ gives $S_f(\lambda)^{1/p} = \varepsilon - \varepsilon r\Psi_p(\mathfrak{D})$, where $(\varepsilon r\Psi_p(\mathfrak{D}))^p = \varepsilon r\Psi_p(\mathfrak{D})$. Raising to the pth power gives the theorem, with $H = \varepsilon$. □

Table 1 lists two exceptional examples. There $S_f(1)$ is computed by direct expansion and $r\Psi_p(\mathfrak{D})$ by Miller chains on the original model (Remark 5.3(b)), independently of Lemma 5.2. The sign in Lemma 5.2 comes from the divisor calculation (B.1); Remark B.1 compares it with the quasi-period formula in Voloch (1997, Section 3) and uses these examples to confirm it.

Table 1: Exceptional examples over $\mathbb{F}_{11}$ with $\lambda = 1$ and origin $(0, 1)$.

$f(w)$	τ	$d = f(1)$	ε	H	$S_f(1)$	$r\Psi_p(\mathfrak{D})$
$1 + w + 3w^3$	1	5	1	1	9	3
$1 + w + 2w^2 + 3w^3$	-1	7	-1	10	0	1

Remark 5.3. (a) Comparing Theorem D with Corollary 4.2 gives

$$r\Psi_{M_\varepsilon}(\mathfrak{D}) = \varepsilon H - 1 + \frac{\mu M_\varepsilon f'(\mu)}{4f(\mu)} + \frac{\mu}{2}\,\alpha_{M_\varepsilon}(R),$$

an identity between a two-pole invariant on E and a one-point Miller coefficient on the twist E_μ. (b) $\Psi_{M_\varepsilon}(\mathfrak{D})$ can also be computed directly: after the model change of Appendix A, $\Psi_{M_\varepsilon}(\mathfrak{D}) = (\alpha_{M_\varepsilon}(R_+) - \alpha_{M_\varepsilon}(R_-))/\kappa_\omega$, where $R_\pm$ are the images of $Q_\pm$ and $\kappa_\omega = -2$ is the scale in (A.4). When $\varepsilon = -1$ the points $R_\pm$ are defined over $\mathbb{F}_{p^2}$. One can run the chain in $\mathbb{F}_p[\xi]/(\xi^2 - f(\mu))$ and split by dynamic evaluation if a zero divisor appears (Della Dora et al., 1985). The twist evaluator of Section 4 avoids this.

6 Binomial applications

Put $a_i = \binom{2i}{i}8^{-i} \in \mathbb{F}_p$ for $i \geq 0$ and $B_K = \sum_{i=0}^{K-1} a_i$ for $K \geq 0$, so $B_0 = 0$. For $0 \leq i \leq h$, $\binom{h}{i} \equiv \binom{-1/2}{i} = (-1/4)^i \binom{2i}{i} \pmod{p}$, since $i!$ is a unit. Hence, for $k \geq 1$,

$$[x^{ki}]\,(1 - x^k/2)^h = a_i \qquad (0 \leq i \leq h), \tag{6.1}$$

and all coefficients of $(1 - x^k/2)^h$ at exponents not divisible by k vanish. The weighted sums

$$U_p(L) = \sum_{i=0}^{L-1} (h + 1 + i)\, t_i \quad (0 \leq L \leq h), \qquad t_0 = 1, \quad t_{i+1} = t_i \,\frac{h + 2 + i}{2i + 2} \quad (0 \leq i < h), \tag{6.2}$$

which motivated this work, reduce to the same data; the denominators $2i + 2 \leq p - 1$ are units. Upper negation gives $t_i \equiv (2i+1)a_i$ for $0 \leq i \leq h$, and the ratio $a_{i+1}/a_i = (2i+1)/(4(i+1))$, valid for $0 \leq i \leq p-2$, yields the telescoping certificate $((2i+1)^2-8)a_i = 4i(2i+5)a_i - 4(i+1)(2i+7)a_{i+1}$, so that

$$U_p(L) \equiv 4B_L - 2L(2L + 5)\, a_L \pmod{p} \qquad (0 \leq L \leq h). \tag{6.3}$$

Each weighted sum therefore needs a prefix B_L and a boundary term a_L.

6.1 The rounded third

Let $p \equiv 2 \pmod 3$, $L = (p + 1)/3$, $m = L - 1$, and $f = 1 - x^3/2$. The exponents $3i \leq p - 1$ are exactly $0 \leq i \leq m$, and $m < h$, so (6.1) gives $T_f(1) = B_L$. The curve $v^2 = 1 - x^3/2$ has j-invariant 0 and is supersingular for $p \equiv 2 \pmod 3$, so $\tau_f = 0$ and $N = p + 1$ on every twist.

Proof of Corollary E. (i) $f(1) = 1/2 \neq 0$, $\varepsilon = \chi(2)$ is read off from $p \bmod 8$, and $N = p + 1$. Corollary 4.2 at $a = 1$ evaluates $T_f(1) = B_L$ with one chain of length $O(\log p)$. No point counting is involved.

(ii) Put $G = m!$. Wilson's theorem in the form $k!\,(p - 1 - k)! \equiv (-1)^{k+1} \pmod{p}$, with $k = 2m$ and $p - 1 - 2m = L$, gives $(2m)!\,L! = -1$. Since $L! = LG$ and $L \equiv 1/3$, this is $(2m)! = -3G^{-1}$. As $3m = p - 2$, we have $8^m = 2^{p-2} = 1/2$, so $a_m = (2m)!/(G^2\,8^m) = -6G^{-3}$. The ratio $a_L/a_m = (2m + 1)/(4L)$ equals $-1/4$, because $2m + 1 \equiv -1/3$ and $4L \equiv 4/3$. Hence $a_L = 3/(2G^3)$. Cubing is a bijection of $\mathbb{F}_p^\times$ when $p \equiv 2 \pmod 3$, with inverse $y \mapsto y^{(2p-1)/3}$, which gives $G = (3/(2a_L))^{(2p-1)/3}$. Finally $B_{L+1} = B_L + a_L$, so either of B_{L+1} and G yields the other in $O(\log p)$ operations once B_L is known. □

Example 6.1 ($p = 23$)**.** Here $L = 8$, $m = 7$, $f(1) = 1/2 \equiv 12$, and $\varepsilon = \chi(2) = 1$ since $23 \equiv 7 \pmod 8$. The normalized polynomial is $g(z) = 2 - (1 + z)^3 = 1 - 3z - 3z^2 - z^3$, so (4.2) gives $E_1 : Y^2 = X^3 + 2$ and $R = (10, 6)$. With $N = 24$ the chain returns $\alpha_{24}(R) = 1$, and (4.5) gives

$$T_f(1) = 1 - \frac{24 \cdot (-3)}{4} - \frac{1}{2} \equiv 1 + 18 - 12 = 7.$$

Directly, $B_8 = 1 + 6 + 8 + 11 + 12 + 10 + 18 + 10 \equiv 7 \pmod{23}$. The next term is $a_8 = 9$, and indeed $7! \equiv 3$ and $3/(2 \cdot 3^3) \equiv 9$.

The boundary also gives the neighbouring cutoff. From $a_m = -4a_L$ and (6.3),

$$U_p(L) = 4B_L - \frac{17}{3}\,G^{-3}, \qquad U_p(m) = 4B_L - \frac{16}{3}\,G^{-3}, \tag{6.4}$$

so $U_p(L) - U_p(m) = -G^{-3}/3 \neq 0$ and the two adjacent sums never vanish together. Fast computation of B_L thus leaves exactly one modular factorial to acquire. The companion report (Jordan, 2026) describes exact methods for that boundary, including Jacobi-sum reconstruction for suitable primes (cf. Tal, 2026) and remainder forests for tables over many primes. It also interprets a_L as a matrix entry, in a specified basis, of the Cartier and Frobenius operators on H^1_{dR} of $y^2 = x^3 + 1/4$; that interpretation is basis-dependent and gives no faster algorithm.

6.2 The quarter point

Let $p = 4n + 1 \geq 17$ and $f = 1 - x^4/2$. The exponents $4i \leq p - 1$ are exactly $0 \leq i \leq n$, and $n \leq h$, so (6.1) gives $T_f(1) = B_n + a_n$ and $H = a_n$. The curve has $j = 1728$, and its trace is classical complex-multiplication data. If $p = u^2 + s^2$ with $u \equiv 1 \pmod 4$, Gauss's congruence $\binom{2n}{n} \equiv 2u \pmod p$ (see Cosgrave & Dilcher, 2010, Theorem 1) gives $a_n \equiv 2u\, 8^{-n}$. Since $\tau_f \equiv H = a_n$ and $2\sqrt{p} < p/2$, the trace is the representative of a_n in $(-p/2, p/2)$. Formula (6.5) below also holds at $p = 13$, where Hasse's bound still leaves the single candidate $\tau_f = 4$ and $B_3 = 5$. Cornacchia's algorithm supplies u. Corollary 4.2 then gives

$$B_n = T_f(1) - a_n, \qquad U_p(n) \equiv 4B_n + \tfrac{9}{4}\, a_n \pmod p. \tag{6.5}$$

For example, at $p = 97$ one finds $n = 24$, $\tau_f = -18$, $a_n = 79$, $B_n = 33$ and $U_p(n) = 43$. Here both the prefix and the boundary are available in time polynomial in $\log p$, with trace preparation separate from the per-query cost. Cornacchia's route needs a square root of -1 modulo p, which is found in expected polynomial time from a quadratic nonresidue (deterministically under GRH); alternatively, Schoof's algorithm computes τ_f unconditionally and deterministically in polynomial time, after which $a_n \equiv \tau_f \pmod p$. The quarter point thus takes its boundary coefficient from classical complex-multiplication data, whereas the examples in Section 6.3 simplify the Miller evaluation itself.

6.3 Torsion points and closed forms

For general inputs the evaluator of Theorem B requires computing $\alpha_N(R)$. When R has small order, the composition law (3.3) reduces this computation to a closed form. Throughout, $h = (p-1)/2$ and $H = [x^{p-1}]f^h$.

Proposition 6.2 (Torsion collapse)**.** *In the setting of Theorem B, suppose that R has order m in $E_a(\mathbb{F}_p)$, and put $q = N/m$, an integer formed before reduction modulo p. Then $\alpha_N(R) = q\,\alpha_m(R)$ and*

$$C_f(a) = \varepsilon\Big(-\frac{g_1}{4}(N-2) - \frac{q}{2}\,\alpha_m(R)\Big), \qquad T_f(a) = \varepsilon\Big(1 - \frac{aNg_1}{4} - \frac{aq}{2}\,\alpha_m(R)\Big) \quad (a \neq 0). \tag{6.6}$$

Here $N \equiv 1 - \varepsilon H \pmod p$; when $p \nmid m$ this also determines $q \equiv m^{-1}(1 - \varepsilon H) \pmod p$.

Proof. Lagrange's theorem gives $m \mid N$, so $q \in \mathbb{Z}$. By (3.3) with $Q = R$, $\alpha_N(R) = q\,\alpha_m(R) + \alpha_q([m]R) = q\,\alpha_m(R)$; equivalently $f_{N,R} = c\, f_{m,R}^{\,q}$ for a nonzero constant c, because $[m]R = O$ makes $f_{q,O}$ constant, and constants do not change logarithmic derivatives. Substituting into (4.3) and (4.5) gives (6.6). Finally $N = p + 1 - \varepsilon\tau_f \equiv 1 - \varepsilon H$ by Lemma 2.1. If $p \mid m$, for instance $m = N = p$ at trace one, the residue of q is not determined by $N \bmod p$; the formula still holds with the integer q. □

For $f = 1 - 4z\,x^k$ the congruence $\binom{h}{i}(-4)^i \equiv \binom{2i}{i}$, valid for $0 \leq i \leq h$ (see before (6.1)), gives $[x^{ki}]f^h = \binom{h}{i}(-4z)^i \equiv \binom{2i}{i} z^i$, so

$$T_f(1) = S_k(z) := \sum_{i=0}^{\lfloor (p-1)/k \rfloor} \binom{2i}{i} z^i \qquad (k \in \{3, 4\}).$$

For $k = 3$, $p > 3$ and $f(1) \neq 0$, the normalized twist at $a = 1$ is a nonsingular curve $E_1 : Y^2 = X^3 + B_g$. In both examples below R has order 3, and (6.6) with $a = 1$ and $m = 3$ reduces to

$$S_3(z) \equiv \varepsilon\Big(1 - \frac{2N}{3}\Big) \equiv \frac{\varepsilon + 2H}{3} \pmod p. \tag{6.7}$$

Theorem 6.3. *For every prime $p > 3$,*

$$\sum_{k=0}^{\lfloor p/3\rfloor} \binom{2k}{k} \equiv \begin{cases} -\dfrac{1}{3} \pmod p, & p \equiv 2 \pmod 3, \\ \dfrac{1-2r}{3} \pmod p, & p \equiv 1 \pmod 3, \end{cases}$$

where, for $p \equiv 1 \pmod 3$, $r, s \in \mathbb{Z}$ satisfy $4p = r^2 + 27s^2$ and $r \equiv 1 \pmod 3$; the integer r is signed and determined by these conditions.

Proof. Take $k = 3$, $z = 1$ and $f = 1-4x^3$, which is squarefree for $p > 3$; the sum is $S_3(1) = T_f(1)$, since $\lfloor p/3\rfloor = \lfloor (p-1)/3\rfloor$. Here $f(1) = -3 \neq 0$, $\varepsilon = \chi(-3)$, which is 1 exactly when $p \equiv 1 \pmod 3$, and $g_1 = f'(1)/f(1) = 4$. The normalized polynomial is $g(z) = 1+4z+4z^2+\frac{4}{3}z^3$, so (4.2) gives $E_1 : Y^2 = X^3 - \frac{16}{27}$ and $R = (\frac{4}{3}, -\frac{4}{3})$. The tangent slope at R is -2 and $[2]R = (\frac{4}{3}, \frac{4}{3}) = -R$; as $R \neq O$, R has exact order 3, and Algorithm 2.2 gives $\alpha_3(R) = -2 + 0 = -2$. Then (6.6) gives $S_3(1) = \varepsilon(1 - N + N/3)$, which is (6.7). If $p \equiv 2 \pmod 3$, then $\varepsilon = -1$ and $H = 0$ because $3 \nmid p-1$, so $S_3(1) \equiv -\frac{1}{3}$. If $p \equiv 1 \pmod 3$, then $\varepsilon = 1$ and, with $L = (p-1)/3$, $H = \binom{h}{L}(-4)^L \equiv \binom{2L}{L} \equiv -r$ by Jacobi's congruence (Berndt, Evans & Williams, 1998; Cosgrave & Dilcher, 2010), so $S_3(1) \equiv (1-2r)/3$. □

Example 6.4 ($z = -2$)**.** For $f = 1 + 8x^3$ one has $f(1) = 9$, $\varepsilon = 1$, $g_1 = \frac{8}{3}$, $E_1 : Y^2 = X^3 + \frac{64}{729}$ and $R = (0, \frac{8}{27})$. The tangent at R is horizontal, so $[2]R = -R$; as $R \neq O$, R has exact order 3, and $\alpha_3(R) = 0$. Then (6.6) gives $S_3(-2) \equiv 1 - \frac{2N}{3}$, which is (6.7) with $\varepsilon = 1$. For $p \equiv 2 \pmod 3$, $H = 0$. For $p \equiv 1 \pmod 3$, $L = (p-1)/3$ is even, so $H = \binom{h}{L}8^L \equiv (-2)^L\binom{2L}{L} \equiv -2^{(p-1)/3}r$, with r as in Theorem 6.3. Therefore, for every prime $p > 3$,

$$\sum_{k=0}^{\lfloor p/3\rfloor} \binom{2k}{k}(-2)^k \equiv \begin{cases} \dfrac{1}{3} \pmod p, & p \equiv 2 \pmod 3, \\ \dfrac{1 - 2\cdot 2^{(p-1)/3}\, r}{3} \pmod p, & p \equiv 1 \pmod 3. \end{cases}$$

Remark 6.5. As discovery provenance only (the proof above establishes the result): Theorem 6.3 was first observed in a pre-registered integer-relation search. That search tested whether $S_k(z)$ is a combination with small rational coefficients of Fermat quotients, CM data and power-residue symbols. It covered $k \in \{3, 4\}$, ten values of z (including $z = \frac{1}{8}$) and all residue classes modulo 24, a total of 224 combinations, and $z = 1$ with $k = 3$ was the only case that produced a relation. Proposition 6.2 explains the relation that was found, and the negative results are consistent with it: over $\mathbb{Q}$, R has order 3 for $(k, z) = (3, 1)$ and $(3, -2)$, whereas for the other 20 combinations of $k \in \{3, 4\}$ with those ten values and $z = -2$ it has no order up to 12, hence infinite order by Mazur's theorem (Mazur, 1977). This does not prove that no closed form exists in those cases. The value $z = -2$ was not part of the search; Example 6.4 was derived from the proposition and then checked. Neither statement has been located in the literature we examined, and both may be known. These special evaluations do not address the general rounded-third boundary problem of Corollary E.

7 Numerical checks and open questions

The accompanying script `cm_verify.py` (pure Python 3, no dependencies) checks each identity against direct computation. Table 2 reports one full run, under CPython 3.13 on Linux x86_64; its output is saved with the script as `cm_verify_output.txt`. All comparisons agreed. Rows marked † use the seed 2026102202, rows marked ‡ (Section 6.3) are deterministic, and the other rows use the seed 20261022. These are finite checks, not proofs. Earlier and larger campaigns, including disjoint confirmation samples, are documented separately in the companion report and are not counted here.

Table 2: Checks executed by `cm_verify.py` in one full run. Random squarefree cubics and quartics over primes $5 \le p < 140$ unless stated.

Statement and sample	Comparisons
Theorem A (3.2): every affine rational point of random short curves, $p < 80$; 54 chains meet $T = P$ at an addition step	4,601
† Normalization of Theorem A: $f_{N,P}$ from a basis of $L(N(O))$, expanded in $t = -x/y$, against the chain and $bJ_W(a)$; all points with $b \neq 0$ on sampled curves, $5 \le p \le 23$, with a trace-one curve where one exists (108 points with $N = p$)	354
Lemma 2.1: every pole, including branch points	21,114
Theorem B (4.1): every nonbranch pole, including 303 at $a = 0$	20,837
Corollary 4.2: every nonzero nonbranch argument	20,534
Theorem D: random arguments, $f(0) = 1$, $p \ge 7$ (69 exceptional)	2,739
Theorem 4.4, both outputs, under $r_i \neq 0 \Rightarrow f(a_i) \neq 0$: 765 queries, $s \le 4$, $p < 80$, 60 excluded branch points of zero residue	1,530
† Theorem 4.4, both outputs, exact $C_f(a_i)$: 448 queries, $1 \le s \le 4$, $p < 80$, 203 active branch poles	896
Identity (4.9): every pole, $p < 60$	2,149
† (6.1) by polynomial powering, the ratio of consecutive a_i, and $t_i \equiv (2i+1)a_i$, for $0 \le i \le h$; all primes $5 \le p < 1000$	166 primes
† Congruence (6.3) against definition (6.2) for every $0 \le L \le h$, same primes	38,144
Corollary E and (6.4): all 221 eligible primes $5 \le p < 3000$; part (i) also at the additional prime 1,000,211	$221 + 1$ primes
Equation (6.5): all 210 eligible primes $13 \le p < 3000$ and the additional prime 1,000,033	$210 + 1$ primes
‡ Section 6.3: order of R over $\mathbb{Q}$ for the 22 pairs (k, z) of Remark 6.5 and $\alpha_3(R)$ for $z = 1, -2$, in exact arithmetic; Theorem 6.3 and Example 6.4 by direct summation at all primes $5 \le p < 3000$	856

Open questions

1. *Branch poles.* At $f(a) = 0$ the two simple poles merge into a pole of the second kind. Lemma 2.1 still holds, but Theorem B does not apply; an $O(\log p)$ evaluation of $J_f(a)$ there would need a further local coefficient. Of the methods in this paper, only the trace-free $\widetilde{O}(\sqrt{p})$ route of Remark 4.5 covers branch poles.
2. *Higher genus.* The residue cancellation of Proposition 5.1 extends formally, with the Hasse invariant replaced by the Cartier–Manin matrix, but evaluation at one point gives only one linear functional of the holomorphic correction. We have not examined whether Miller-type chains on the Jacobian evaluate truncations of $f^{(p-1)/2}$ for $\deg f \ge 5$.
3. *Extension fields and higher powers of p.* Proposition 5.1 holds over $\mathbb{F}_{p^k}$; the base-field simplification and the twist evaluator would need separate arguments there, which we have not given. All results here are modulo p; we have not examined lifts modulo p^2.
4. *Torsion families.* Which (k, z), or more generally which (f, a), make R a torsion point, and do the resulting closed forms account for known congruences for truncated central-binomial sums? We have examined only the cases in Remark 6.5.
5. *Boundary terms.* For $p \equiv 2 \pmod 3$, is there a polynomial-time algorithm for $((p-2)/3)! \bmod p$, or equivalently, by Corollary E, for $B_{(p+4)/3}$? The methods discussed here do not provide one.

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the content rests with the author.

A Cubic and quartic model changes

Let $f = 1 + f_1 w + f_2 w^2 + f_3 w^3 + f_4 w^4$, with $f_4 = 0$ for a cubic, and take $(0, 1)$ as origin (cf. Connell, 1999, Section 1.2). Set

$$x = \frac{2(v+1) + f_1 w}{w^2}, \qquad y = \frac{4(v+1) + 2(f_1 w + f_2 w^2) - f_1^2 w^2/2}{w^3}. \tag{A.1}$$

Then $y^2 + a_1 xy + a_3 y = x^3 + a_2 x^2 + a_4 x + a_6$ with

$$a_1 = f_1, \quad a_2 = f_2 - f_1^2/4, \quad a_3 = 2f_3, \quad a_4 = -4f_4, \quad a_6 = a_2 a_4. \tag{A.2}$$

Near $(0, 1)$ one has $x = 4w^{-2} + 2f_1 w^{-1} + O(1)$ and $y = 8w^{-3} + O(w^{-2})$, so $(0, 1)$ maps to O. With $b_2 = a_1^2 + 4a_2$, $b_4 = 2a_4 + a_1 a_3$, $b_6 = a_3^2 + 4a_6$, the substitution $X = x + b_2/12$, $Y = y + (a_1 x + a_3)/2$ gives $Y^2 = X^3 + AX + B$ with

$$A = \frac{b_4}{2} - \frac{b_2^2}{48}, \qquad B = \frac{b_6}{4} - \frac{b_2 b_4}{24} + \frac{b_2^3}{864}. \tag{A.3}$$

All denominators are units for $p \geq 5$. Comparing leading terms at O fixes the differential scale

$$\frac{dw}{v} = \kappa_\omega \frac{dX}{2Y}, \qquad \kappa_\omega = -2. \tag{A.4}$$

For $f = g$ as in (4.1), these formulas give (4.2), and the identity (4.4) follows from (A.1) by direct substitution.

B Proof of the exceptional normalization

We keep the notation of Section 5: $\tau = \tau_f = \varepsilon \in \{\pm 1\}$, $k = \overline{\mathbb{F}}_p$, $\omega = dw/v$, Ψ_p as in (5.3), and σ coefficient conjugation.

Proof of Lemma 5.2. Step 1: kernel of Verschiebung. We have $\pi + \mathrm{Ver} = [\varepsilon]$, $\pi \,\mathrm{Ver} = [p]$, $\mathrm{Ver}^* \omega = \varepsilon\omega$ and $\mathcal{C}\omega = \varepsilon\omega$. The curve is ordinary, so Ver is separable of degree p, and $K = \ker \mathrm{Ver}(k)$ is the cyclic group of geometric p-torsion; every $T \in K$ satisfies $\pi(T) = \varepsilon T$. The divisor $\Delta = \sum_{T \in K} (T) - p(O)$ is principal and Galois-stable, so $\Delta = \mathrm{div}(s)$ for some s defined over $\mathbb{F}_p$. Put $\xi = d \log s$ and $\psi(T) = \Psi_p([(T) - (O)])$. Writing t_T for translation by T, permutation of K gives

$$\mathrm{div}\, \frac{t_T^* s}{s} = p(O) - p(-T), \qquad t_T^* \xi - \xi = \psi(T)\, \omega, \tag{B.1}$$

with positive sign because $[(O) - (-T)] = [(T) - (O)]$. Moreover $\mathcal{C}\xi = \xi$ and $\psi(T)^p = \varepsilon \psi(T)$.

Step 2: descent through Ver. For $R \in E(k)$, (B.1) makes $t_{-R}^* \xi - \xi$ invariant under K, so it descends through the separable quotient Ver: there is a unique differential Θ_R with $\mathrm{Ver}^* \Theta_R = t_{-R}^* \xi - \xi = d \log(t_{-R}^* s / s)$. As Ver is étale, Θ_R has only simple poles, with residue divisor $(\mathrm{Ver}\, R) - (O)$. Cartier commutes with separable pullback, since a separating coordinate stays separating, and separable pullback is injective, so $\mathcal{C}\Theta_R = \Theta_R$. Translation and conjugation give

$$\Theta_{R+T} = \Theta_R - \varepsilon \psi(T)\, \omega \quad (T \in K), \qquad \sigma(\Theta_R) = \Theta_{\pi(R)}. \tag{B.2}$$

Step 3: generators. Let $P \neq O$ be geometric p-torsion and choose R with $\mathrm{Ver}\, R = P$, so that $\pi(R) = \varepsilon R - P$. For $\varepsilon = 1$ put $\zeta = \Theta_R$, $\mathfrak{D}_P = (P) - (O)$ and $\beta = \psi(P)$. For $\varepsilon = -1$ put $\zeta = \Theta_R - \Theta_{-R}$, $\mathfrak{D}_P = (P) - (-P)$ and $\beta = 2\psi(P)$; the factor 2 appears because

$[(P)-(-P)] = 2[(P)-(O)]$. In both cases ζ has residue divisor $\mathfrak{D}_P$, $\beta = \Psi_p([\mathfrak{D}_P])$, and by (B.2)

$$\mathcal{C}\zeta = \zeta, \qquad \sigma(\zeta) = \varepsilon\zeta + \varepsilon\beta\,\omega, \qquad \beta^p = \varepsilon\beta.$$

A differential θ_P with the same residues and $\sigma(\theta_P) = \varepsilon\theta_P$ differs from ζ by $c\,\omega$, where the eigencondition forces $c^p - \varepsilon c = -\varepsilon\beta$, that is $\varepsilon c^{1/p} - c = \beta$. Hence $(\mathcal{C}-1)\theta_P = (\mathcal{C}-1)(c\,\omega) = \varepsilon c^{1/p}\omega - c\,\omega = \beta\,\omega$.

Step 4: general divisors. Represent the class of $\mathfrak{D}_0$ by some $\mathfrak{D}_P$, taking half the class when $\varepsilon = -1$, which is possible because 2 is invertible on a group of order p; the zero class is handled by $d\log u$ directly. Write $\mathfrak{D}_0 - \mathfrak{D}_P = \operatorname{div}(u)$. The eigenconditions imply $u^\sigma = c\,u^\varepsilon$ for a constant c, so $d\log u$ is an eigen-differential fixed by $\mathcal{C}$. The difference $\theta_{\mathfrak{D}_0} - \theta_P - d\log u = b\,\omega$ is holomorphic with $b^p = \varepsilon b$, hence also fixed by $\mathcal{C}$. Therefore $(\mathcal{C}-1)\theta_{\mathfrak{D}_0} = (\mathcal{C}-1)\theta_P = \Psi_p([\mathfrak{D}_P])\,\omega = \Psi_p([\mathfrak{D}_0])\,\omega$. □

Remark B.1 (Comparison with Voloch's quasi-periods)**.** Voloch (1997) defines quasi-periods on an ordinary curve by $z(P+P_0) = z(P) + \eta(P_0)$ for $P_0 \in \ker V$, with z normalized so that $z + y/x$ vanishes at O and $\omega = dx/(2y)$, and in Section 3 writes $\eta(P) = -df/(f\omega)$ for f with divisor $p((P)-(O))$. Write ω' for Voloch's differential; ours is $\omega = dw/v$, and the two differ by a constant factor, cf. (A.4). Both $z\omega'$ and ξ have simple poles with residue 1 at every point of K and no other poles, so $z = \xi/\omega' + \text{const}$. Then (B.1) gives $\eta(T) = +dF/(F\omega')$ for $\operatorname{div} F = p((T)-(O))$, the opposite sign to the printed formula; the constant factor between ω and ω' does not affect this sign. We have not determined whether this reflects a convention we have not matched or a misprint, and our argument does not use Voloch's formula. The sign matters: replacing ψ by $-\psi$ in (B.1) reverses β in Step 3 and would give $S_f(\lambda) = H + \varepsilon r\Psi_p(\mathfrak{D})$ in the exceptional case. For the two rows of Table 1 that alternative gives 4 and 9, whereas the directly computed values are 9 and 0, as Theorem D states. The 69 exceptional cases in Table 2 agree with Theorem D as well.

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